



微积分第二次作业

3. (4) $\lim_{n \rightarrow \infty} \frac{n^2}{2^n} = 0$

证明: 对于 $\forall \varepsilon > 0$, 找 $|\frac{N^2}{2^N} - 0| = \frac{N^2}{2^N} < \varepsilon$

~~$$2^N = (1+1)^N = 1 + C_N^1 + C_N^2 + \dots + C_N^N$$~~

$$\text{令 } f(x) = 2^x - \frac{x^2}{6}, \quad f'(x) = 2^x \ln 2 - \frac{x}{3}, \quad f''(x) = 2^x (\ln 2)^2 - \frac{1}{3}, \quad f'''(x) = 2^x (\ln 2)^3 - 1$$

$x \geq 4$ 时, $f'''(x) > 0$, $f''(x)$ 增, 故易知 $f''(x) > 0$ ($x \geq 4$ 时), 同理 $f'(x) > 0$ ($x \geq 4$), $f(x)$ 增

$$\therefore f(4) = 2^4 - \frac{4^2}{6} = \frac{16}{3} > 0 \quad \therefore f(x) > 0 \quad (x \geq 4) \quad \therefore N \text{ 足够大时, } 2^N > \frac{N^2}{6}$$

$$\therefore \frac{N^2}{2^N} < \frac{N^2}{N^2/6}, \text{ 只需让 } \frac{N^2}{N^2/6} = \frac{6}{N} < \varepsilon, \text{ 令 } N = \left\lceil \frac{6}{\varepsilon} \right\rceil + 1 \text{ 即可.}$$

当 $N \rightarrow n > N = \left\lceil \frac{6}{\varepsilon} \right\rceil + 1$ 时, 恒有 $|\frac{n^2}{2^n} - 0| < \varepsilon$, 成立。□

3. (6) $\lim_{n \rightarrow \infty} n^2 q^n = 0 \quad (|q| < 1)$

证明: 令 $t = \frac{1}{|q|}$, 则 $|q^n| = \frac{1}{t^n}$, $|n^2 q^n| = \frac{n^2}{t^n} \quad (t > 1)$

$$\text{令 } f(x) = t^x - kx^2, \text{ 同 3. (4) 可知 } f'''(x) = t^x (\ln t)^3 - 2k, \text{ 令 } f'''(x) \geq 0,$$

$$\text{则有 } t^x (\ln t)^3 = 2k, \quad t^x = \frac{2k}{(\ln t)^3}, \quad x = \log_t \frac{2k}{(\ln t)^3}, \text{ 只需让 } N > \log_t \frac{2k}{(\ln t)^3}$$

$$\text{不妨令 } N_1 = \left\lceil \log_t \frac{2k}{(\ln t)^3} \right\rceil + 1, \text{ 同 3. (4) 知可得 } x > N_1 \text{ 时 } t^x > kx^2 \quad (k > 0)$$

$$\therefore \frac{N^2}{t^N} < \frac{N^2}{kN^2} = \frac{1}{kN} < \varepsilon, \text{ 只需让 } \frac{1}{kN} < \varepsilon, \text{ 令 } N_2 = \left\lceil \frac{1}{k\varepsilon} \right\rceil + 1$$

当 $n > N = \max\{N_1, N_2\}$ 时, 恒有 $|\frac{n^2}{t^n} - 0| < \varepsilon$, 即 $|n^2 q^n - 0| < \varepsilon$, 得证。□

(理论上当然可以通过调整 k 来把 N_1 调小到 1, 但此处我没有展开处理)

3. (7) $\lim_{n \rightarrow \infty} \left(\frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \dots + \frac{1}{(3n-2)(3n+1)} \right) = \frac{1}{3}$

证明: $\frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \dots + \frac{1}{(3n-2)(3n+1)}$

$$= \left(1 - \frac{1}{4} + \frac{1}{4} - \frac{1}{7} + \dots + \frac{1}{3n-2} - \frac{1}{3n+1} \right) \times \frac{1}{3}$$

$$= \frac{1}{3} - \frac{1}{3} \cdot \frac{1}{3n+1}$$

$$\left| \frac{1}{3} - \frac{1}{3} \frac{1}{3n+1} - \frac{1}{3} \right| < \varepsilon \Rightarrow \frac{1}{3} \left| \frac{1}{3n+1} \right| < \varepsilon, \quad n > \frac{1}{3} \left(\frac{1}{3\varepsilon} - 1 \right) \text{ 即可}$$

$$\text{令 } N_1 = \left\lceil \frac{1}{9\varepsilon} - \frac{1}{3} \right\rceil + 1, \text{ 取 } n > \max\{N_1, 1\} \text{ 即可, 得证。} \quad \square$$

$$3. (8) \lim_{n \rightarrow \infty} \left(\frac{1}{(n+1)^{1/3}} + \frac{1}{(n+2)^{1/3}} + \dots + \frac{1}{(n+n)^{1/3}} \right) = 0$$

证明: $n+1 > n, n+2 > n, \dots, n+n > n$

$$\therefore \frac{1}{(n+1)^{1/3}} + \frac{1}{(n+2)^{1/3}} + \dots + \frac{1}{(n+n)^{1/3}} < \left(\frac{1}{n}\right)^{1/3} \times n = \frac{1}{n^{2/3}}$$

记原式为 t_n , 则 $\frac{1}{n^{2/3}} < \varepsilon$ 时, $|t_n - 0| < \frac{1}{n^{2/3}} < \varepsilon$

$\therefore n^{2/3} > \frac{1}{\varepsilon}, n > \varepsilon^3$, 即让 $N = [\varepsilon^3] + 1$, 则有:

$n > N$ 时, 恒有 $|t_n - 0| < \varepsilon$, 即原命题得证。 $\square$

* 5. 证明: ① A 有限

$$\lim_{n \rightarrow \infty} x_n = A \Leftrightarrow \forall \varepsilon > 0, \exists N, \forall n > N, |x_n - A| < \varepsilon$$

先固定下 ε , 根据 ε 再固定一个 N_1 , 使 $n > N_1$ 时恒有 $|x_n - A| < \varepsilon$

$$\text{则 } \lim_{n \rightarrow \infty} \frac{1}{n} (x_1 + x_2 + \dots + x_n) = \frac{1}{n} \lim_{n \rightarrow \infty} \frac{1}{n} (x_1 + \dots + x_{N_1}) + \lim_{n \rightarrow \infty} \frac{1}{n} (x_{N_1+1} + \dots + x_n) \quad (*)$$

$$n \rightarrow \infty \text{ 时, } \frac{1}{n} (x_1 + \dots + x_{N_1}) \rightarrow 0, \text{ 故 } (*) = \lim_{n \rightarrow \infty} \frac{1}{n} (x_{N_1+1} + \dots + x_n) \quad (**)$$

$$\therefore n > N_1 \text{ 时, } |x_n - A| < \varepsilon$$

$$\therefore |(**) - A| \leq (|x_{N_1+1} - A| + \dots + |x_n - A|) \times \frac{1}{n} < \frac{n - N_1}{n} \varepsilon < \varepsilon$$

$$\therefore \text{即 } \lim_{n \rightarrow \infty} \frac{1}{n} (x_1 + x_2 + \dots + x_n) = A$$

② $A = \pm \infty$, 不妨只考虑正无穷时 (负无穷时方法同理)

对 $\forall G, \exists N_2, n > N_2, x_n > G$

$$\text{即拆成两部分: } \lim_{n \rightarrow \infty} \frac{1}{n} (x_1 + \dots + x_n) = \lim_{n \rightarrow \infty} \frac{1}{n} (x_1 + \dots + x_{N_2}) + \lim_{n \rightarrow \infty} \frac{1}{n} (x_{N_2+1} + \dots + x_n)$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} (x_1 + \dots + x_{N_2}) = 0, \lim_{n \rightarrow \infty} \frac{1}{n} (x_{N_2+1} + \dots + x_n) > \frac{n - N_2}{n} G$$

$$\therefore \lim_{n \rightarrow \infty} \frac{1}{n} (x_{N_2+1} + \dots + x_n) > G$$

$$\text{即 } \forall G, \exists N, n > N \text{ 时, } \frac{1}{n} (x_1 + \dots + x_n) > G$$

$$\text{即 } \lim_{n \rightarrow \infty} \frac{1}{n} (x_1 + x_2 + \dots + x_n) = \infty \quad (-\infty \text{ 情况同理})$$

6. 证明: 同 5 知, $\lim_{n \rightarrow \infty} \frac{1}{n} (x_1 + x_2 + \dots + x_n) = A$

$$\text{即 } \left| \frac{1}{n} (x_1 + x_2 + \dots + x_n) - A \right| < \varepsilon, \forall n > N, \forall \varepsilon$$

则当 $n > N$ 时, $|\sqrt[n]{x_1 x_2 \dots x_n} - A| < \left| \frac{1}{n} (x_1 + x_2 + \dots + x_n) - A \right| < \varepsilon$, 得证。

(突然意识到刚刚那种方法只能处理 x_n 递减的情况, 即 $x_1 > \sqrt{x_1 x_2} > x_2$, 要不然无法保证 $\sqrt{x_1 \cdots x_n} > A$ 。但其他情况我不会证。)

证明: 由已知有结论, $\lim_{n \rightarrow \infty} x_n = A$ 时, $\lim_{n \rightarrow \infty} \sqrt[n]{x_1 x_2 \cdots x_n} = A$

$$\therefore \lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n} = A \quad \therefore \lim_{n \rightarrow \infty} \frac{x_n}{x_{n-1}} = A$$

构造数列 $\{y_n\}$, $y_1 = x_1$, $y_2 = \frac{x_2}{x_1}$, $\dots$, $y_n = \frac{x_n}{x_{n-1}}$, 则有 $\lim_{n \rightarrow \infty} y_n = A$

$$\therefore \lim_{n \rightarrow \infty} \sqrt[n]{y_1 y_2 \cdots y_n} = A, \text{ 即 } \lim_{n \rightarrow \infty} \sqrt[n]{x_n} = A$$